Bibek Kumar Tamang

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Summary

Software Engineer with hands-on experience designing and shipping production-grade backend systems using Java, Spring Boot, and AWS. Proven track record of building scalable microservices, optimizing high-traffic databases, and delivering secure cloud-deployed APIs in Agile environments. Adept at owning features end-to-end — from schema design to containerized deployment — with a growing focus on distributed systems and event-driven architecture.

Skills

- **Programming Language:** Java | SQL | JavaScript | C++ | Python
- **Cloud & DevOps:** AWS | Docker | Git | Kubernetes | CI/CD | Linux | Terraform
- **Backend & Frameworks:** Spring Boot | REST APIs | JWT Authentication | Microservices | React
- **Databases & Messaging:** MySQL | PostgreSQL | MongoDB | Redis | Kafka
- **Observability & Testing:** Junit | Postman | k6 | Log-based Monitoring | Structured Logging
- **Practices:** System Design | Agile/Scrum | Code Review | Database Indexing & Query Optimization

Experience

Dallo Tech — **Jan 2024 - Jun 2025**
Software Developer, Intern — *Lalitpur, Nepal*

- Architected and delivered **12+ production-grade REST API endpoints** for a multi-vendor e-commerce platform serving 3 vendor tiers (admin, seller, customer), reducing average API response time by **35%** through strategic indexing and query optimization on PostgreSQL schemas with 15+ normalized tables.
- Designed and implemented a JWT-based authentication system with role-based access control (RBAC) across 3 user roles, eliminating unauthorized endpoint access and cutting authentication-related support tickets by 40% within the first month of deployment.
- Optimized critical SQL queries on high-traffic product catalog and order tables, introducing composite indexes and query caching strategies that improved read performance by **50%** under concurrent load testing with 200+ simulated users.
- Led backend containerization initiative using Docker, deploying **4 microservices** to AWS EC2 and integrating AWS S3 for static asset delivery, reducing deployment time from **45 minutes to under 8 minutes** via streamlined CI/CD workflows.
- Drove code quality improvements across a team of 5 engineers by participating in **20+ pull request reviews**, refactoring 3 legacy modules to adhere to SOLID principles, and contributing to sprint velocity gains of **~15%** over a 6-week Agile cycle.

Selective Projects

Async Notification Dispatch Engine

Spring Boot | Kafka | SendGrid/Twilio API | PostgreSQL | Docker

- Built an event-driven notification service handling email, SMS, and push delivery using Kafka as the message broker — decoupling 6 upstream producer services and eliminating direct notification coupling from core business logic.
- Implemented idempotent consumer logic with a **deduplication table in PostgreSQL**, reducing duplicate delivery incidents by **100%** across 50,000+ test notification events during integration testing.
- Designed a retry-with-backoff strategy using Kafka dead-letter topics, recovering **97% of failed deliveries** within 3 retry attempts and surfacing unresolved failures to a monitoring dashboard in real time.
- Reduced notification delivery latency from an average of **4.2 seconds to under 800ms** by replacing a synchronous email service call with async Kafka consumption and parallel SendGrid/Twilio dispatch.

Intelligent Log Aggregator with Anomaly Alerting

Spring Boot | Kafka | Elasticsearch | Python | Docker Compose

- Developed a lightweight log ingestion pipeline that collected structured logs from **4 microservices**, streamed them through Kafka, and indexed them into Elasticsearch — enabling full-text search across **1M+ log records** in under 300ms.
- Implemented a statistical anomaly detection layer using Python (Z-score + rolling window) that flagged error-rate spikes with **91% precision**, triggering Slack webhook alerts within 10 seconds of threshold breaches.
- Reduced mean time to detect (MTTD) production incidents from **~12 minutes to under 90 seconds** by replacing manual log grep workflows with real-time structured search and automated alerting.
- Containerized all pipeline components using Docker Compose with volume-mounted configs, enabling full local environment spin-up in **under 3 minutes** for any new team member.

Education

Youngstown State University — **May 2026**
Bachelor of Science, Computer Science

- **Coursework:** Data Structure & Algorithm, Operating System, Object Oriented Programming, Data Science & Machine Learning, Computer Architecture, Networking Concepts & Administration, Development of Database,
- Calculus I, Calculus II, Calculus III, Probability & Statistics, Linear Algebra & Matrix Theory